

Notes: Ali, Lewis, and Vasserman (RES, 2023)

Voluntary Disclosure and Personalized Pricing

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1 Big picture

- **Question.** Do consumers benefit from personalized pricing when they control what information about themselves is disclosed to firms?
- **Motivation.** Firms increasingly infer willingness to pay from consumer data and use those data for targeted ads, discounts, and price discrimination. Privacy regulation often gives consumers consent rights, but equilibrium pricing depends on what firms infer both from disclosure and from silence.
- **Core tension.** Consumer control sounds protective, but disclosure games can unravel: if low types disclose to get lower prices, non-disclosure may itself become a bad signal. The paper studies when control helps rather than hurts.
- **Central distinction.** The effect depends on two forces: market structure and disclosure technology. Monopoly requires richer or group-based evidence to help consumers; competition can make even simple evidence valuable.
- **Monopoly result.** With simple evidence, voluntary disclosure does not improve consumer welfare. With rich evidence, consumers can form segments that induce lower group prices, including the Zeno partition construction.
- **Competition result.** With two horizontally differentiated firms, simple track/do-not-track evidence can improve consumer welfare because consumers can reveal to the distant firm and conceal from the close firm, intensifying competition.
- **Mechanism.** Disclosure is not only a way to tell a seller one's value. It is also a strategic tool for changing the seller's residual demand, creating or amplifying competitive outside options.
- **Policy relevance.** Opt-in/opt-out design, data portability, third-party verification, and the ability to selectively disclose to different sellers matter because they determine the feasible messages in the disclosure game.
- **Main takeaway.** Data control does not automatically protect consumers. It helps when the disclosure technology enables welfare-improving pooling or when consumer-controlled disclosure strengthens competition among firms.

2 Incremental contribution of the paper

- **Contribution relative to standard privacy debates.** Many privacy discussions treat less disclosure as better for consumers because firms use data to price discriminate. This paper shows that the welfare effect is equilibrium-dependent: controlled disclosure can lower prices and raise consumer surplus.
- **Contribution relative to voluntary disclosure/unravelling theory.** Classical disclosure models often emphasize that verifiable disclosure unravels or that non-disclosure is interpreted pessimistically. Ali, Lewis, and Vasserman show that unravelling logic is incomplete once prices are endogenous and disclosure can generate market segments.
- **Contribution to price discrimination theory.** The paper separates personalized pricing into two components: information acquisition by firms and strategic disclosure by consumers. This distinction explains why personalized pricing can either extract surplus or create discounts.
- **Contribution on disclosure technology.** The paper provides a taxonomy: simple evidence, rich evidence, group-pricing evidence, opt-in/opt-out policies, and selective disclosure across firms. These technologies are not equivalent; the welfare implications depend on which messages consumers can credibly send.
- **Contribution on competition.** The competitive-market analysis shows that consumer data can be used to intensify competition rather than merely to enable price discrimination. Consumers exploit the fact that each firm wants to poach them from rivals.

- **Contribution to platform and intermediary design.** The examples in Section 5 translate the theory into verification institutions: governments, platforms, and firms can act as administrators that decide what evidence consumers can use.
- **Conceptual novelty.** The paper reframes privacy as *control over verifiable disclosure*, not simply secrecy. The key object for policy is the feasible message space and whether consumers can use it to create favorable market segmentation.

3 Model objects and measurement

3.1 Consumer type and willingness to pay

In the monopoly sections, the consumer has a private valuation

$$v \in [\underline{v}, \bar{v}],$$

drawn from a distribution F . A monopolist sells one unit of a good. The consumer can sometimes send verifiable evidence about v before the seller chooses a price.

Interpretation: The model is about hard information. The consumer cannot lie about evidence, but she can choose which evidence to disclose if her evidence technology permits it.

3.2 Evidence technology

An evidence technology specifies the set of messages available to each type. A message is credible only if the true type belongs to the set represented by the message.

- **Simple evidence:** the consumer can fully reveal her type or disclose nothing.
- **Rich evidence:** the consumer can disclose arbitrary subsets/intervals that contain her type, enabling partial disclosure.
- **Group-pricing evidence:** some low-value consumers can send a verifiable group message that high-value consumers cannot send.
- **Selective evidence in competition:** the consumer can disclose different messages to different firms.

Interpretation: The evidence technology is the paper’s main policy object. Consent rules, opt-in defaults, platform dashboards, government verification, and tracking programs all implement different feasible message spaces.

3.3 Seller beliefs and pricing

After receiving a message m , the seller forms posterior beliefs over the consumer’s type. In a monopoly segment m , the seller chooses a price that maximizes expected revenue:

$$p(m) \in \arg \max_p p \cdot \Pr(v \geq p \mid m).$$

If a segment is constructed so that the seller prices at its lower bound, all consumers in the segment buy and higher types in the segment earn surplus.

Interpretation: Pricing is the force that transforms information disclosure into welfare effects. A disclosure message is useful to consumers if it induces a price below what they would otherwise pay.

3.4 Consumer welfare objects

The paper distinguishes:

- **interim consumer surplus:** the payoff of a specific type after her type is realized;
- **ex ante consumer surplus:** expected consumer surplus before type is realized;
- **Pareto improvement:** every consumer type is weakly better off, and some are strictly better off;
- **producer surplus:** the seller's expected revenue/profit under a disclosure equilibrium.

Interpretation: The paper often emphasizes interim improvements because privacy policy is distributional: some consumers can be hurt even if average surplus rises.

3.5 Competitive market objects

In the competitive sections, two firms are horizontally differentiated:

$$\ell_L = -1, \quad \ell_R = 1, \quad \ell \in [-1, 1].$$

The consumer's location ℓ is private. If she buys from firm i , she pays price p_i and transportation cost $|\ell_i - \ell|$.

Interpretation: Location is a metaphor for product fit, brand preference, or switching/search cost. Disclosure can reveal who is close to the consumer and therefore who has market power.

4 Models and theoretical specifications

4.1 Benchmark monopoly without disclosure

A monopolist with no personalized information chooses a uniform price

$$p^* \in \arg \max_p p[1 - F(p)].$$

Consumers with $v \geq p^*$ buy. Consumers below p^* do not buy.

Interpretation: Uniform pricing pools all types. It protects high-value consumers from individualized extraction but excludes some low-value consumers who would buy at a lower price.

4.2 Simple evidence under monopoly

With simple evidence, the consumer can either reveal v exactly or send the uninformative message. The paper's negative result is:

Proposition 1. With simple evidence, across all equilibria, the consumer's interim payoff is not improved relative to the no-disclosure benchmark.

Interpretation: Full revelation to a monopolist leads to personalized prices that extract surplus from revealed types. Non-disclosure is also informative in equilibrium, so consumers cannot use silence to generate a better pooled price.

4.3 Rich evidence and Zeno's partition

With rich evidence, consumers can send partial messages. For a uniform distribution on $[0, 1]$, one equilibrium partitions types into intervals:

$$m_0 = [1/2, 1], \quad m_1 = [1/4, 1/2], \quad m_2 = [1/8, 1/4], \quad \dots$$

Each segment is priced at its lower bound.

For example, in segment $[1/2, 1]$, the monopolist charges $1/2$, and all consumers in that segment purchase. The same logic repeats on successively smaller intervals.

Proposition 2. With rich evidence, there is a Pareto-improving equilibrium in which almost all consumer types are better off than under uniform pricing.

Interpretation: Rich evidence lets consumers create favorable pooling. The monopolist still price discriminates across segments, but each segment contains high types who receive surplus because the segment price equals the lower bound.

4.4 Efficient partitional equilibria

The paper defines an efficient partitional equilibrium as one in which types are partitioned into connected market segments and, within each segment, the seller charges the lowest type in that segment.

Proposition 3. Given any equilibrium, there is an efficient partitional equilibrium that results in the same interim consumer surplus for almost every type.

Interpretation: This result justifies focusing on partitions. It says that complicated disclosure strategies can be replaced by payoff-equivalent segmentation structures, making the economics transparent.

4.5 Optimality of greedy segmentation for power distributions

For $[0, 1]$ and $F(v) = v^k$, the paper shows:

Proposition 4. The greedy segmentation is the consumer-optimal equilibrium segmentation.

Interpretation: The greedy/Zeno-style construction is not only illustrative for this class of distributions; it is ex ante consumer optimal. The result clarifies when the simple iterative construction is not just a feasible improvement but the best equilibrium segmentation for consumers.

4.6 Partial disclosure and group pricing

The evidence technology facilitates group pricing if some message can be sent by a positive mass of low-value consumers and cannot be sent by sufficiently high-value consumers. Then:

Proposition 5. If an evidence technology facilitates group pricing, then there is a Pareto-improving equilibrium.

Interpretation: Full rich evidence is not necessary. A student ID, EBT card, senior-citizen card, or other group-verification device can create a discount group without requiring consumers to reveal exact willingness to pay.

4.7 Competition: full revelation benchmark

With two firms, full revelation allows firms to know the consumer's location. In competitive equilibrium, this can increase competition for centrally located consumers because both firms can profitably serve them.

Proposition 6. There is a fully revealing equilibrium in both simple and rich evidence games.

Interpretation: Full revelation is not necessarily bad under competition. Unlike monopoly, revelation may reduce prices by making both firms compete for the consumer.

4.8 Competition with simple evidence

The key competitive construction is selective disclosure. Centrally located consumers reveal to both firms. Extreme types reveal location only to the distant firm and conceal it from the closer firm.

Proposition 7. With simple evidence, there is an equilibrium in which the consumer's disclosure strategy improves payoffs by amplifying competition.

Interpretation: Consumers strategically disclose to create an outside option. The distant firm sees enough to poach; the close firm faces competitive pressure because non-disclosure indicates the consumer might be poachable.

4.9 Competition with rich evidence

Rich evidence creates a more refined segmentation in the competitive setting. The construction resembles Zeno's partition but is adapted to two-sided horizontal differentiation.

Proposition 8. With rich evidence, there is an equilibrium with segmented disclosure strategies and lower prices for consumers.

Interpretation: Rich evidence can refine price competition, but the main competitive insight is already present with simple evidence: selective disclosure can be enough to force better prices.

4.10 Comparison with uniform pricing

For symmetric log-concave densities, the paper compares disclosure equilibria with the benchmark symmetric uniform-price equilibrium.

Proposition 9. Every consumer type has a strictly higher payoff in the constructed simple- and rich-evidence equilibria than in the uniform-pricing benchmark.

Proposition 10. With simple and rich evidence, every efficient passive equilibrium has higher average consumer surplus than the benchmark with uniform pricing.

Interpretation: These propositions establish that the positive competitive result is not limited to one special equilibrium. Within a natural class, voluntary disclosure improves average consumer welfare relative to uniform pricing.

5 Identification and theoretical design logic

5.1 Why the paper is theoretical rather than empirical

The paper does not estimate treatment effects. Its contribution is conceptual: it characterizes equilibria of disclosure games and identifies when consumer control over data improves welfare.

5.2 What disciplines the results

- **Bayesian inference:** firms infer types from both disclosed messages and non-disclosure.
- **Sequential optimality:** sellers must set optimal prices given posterior beliefs.
- **Verifiability:** consumers cannot send messages inconsistent with their true type.
- **Equilibrium incentives:** consumers must prefer their prescribed disclosure messages.
- **Welfare comparisons:** the paper compares interim and ex ante consumer payoffs across disclosure technologies and market structures.

Interpretation: The results are disciplined by equilibrium restrictions rather than econometric identification. The logic is about which prices and beliefs can be sustained when disclosure is verifiable but voluntary.

5.3 Why simple and rich evidence are the key design contrast

Simple evidence captures track/do-not-track or full opt-in/opt-out rules. Rich evidence captures finer consent dashboards or verification systems that allow consumers to disclose membership in ranges/groups. The contrast shows why “consumer control” is too broad a policy label: the exact menu of messages is decisive.

5.4 Why market structure matters

A monopolist uses information to extract surplus. Competing firms use information to poach consumers from rivals. This difference flips the welfare effect of disclosure. The same disclosure option that hurts or fails to help under monopoly can help under competition.

5.5 Limitations of the theoretical design

- The paper abstracts from firms acquiring data without consumer permission.
- It does not model endogenous investment in data collection, ad targeting, or platform design.
- Many results assume verifiable hard information, not cheap talk.
- The competitive model is stylized with full coverage and horizontal differentiation.
- The paper focuses on consumer surplus and producer surplus, not privacy utility from secrecy itself.

Interpretation: These limitations do not weaken the main mechanism; they clarify what the paper isolates: how verifiable consumer-controlled disclosure changes equilibrium prices.

6 Section-by-section study guide

6.1 Section 1: Introduction

The introduction frames the paper around the policy debate over privacy, data control, and personalized pricing. It emphasizes that firms infer from both disclosure and non-disclosure, so giving consumers control over data has subtle equilibrium effects.

6.2 Section 2: Setup

The model introduces consumer types, evidence technologies, seller beliefs, and pricing. The key modeling choice is to treat data as verifiable evidence that consumers can choose to reveal partially, fully, or not at all.

6.3 Section 3: Monopoly

This section contains the main contrast between simple evidence and rich evidence. Simple evidence cannot improve welfare under monopoly, but rich evidence enables segment pricing that can improve consumer surplus. Group-pricing evidence offers a coarser but practically relevant route to the same kind of improvement.

6.4 Section 4: Competition

This section shows that competition changes the logic of disclosure. Consumers can reveal information to some firms and conceal from others. Selective disclosure makes firms compete harder and lowers consumer costs.

6.5 Section 5: Practical mechanisms

The paper discusses government verification, intermediary/platform verification, and firm-run verification. Examples include EBT discounts, platform ad preference dashboards, and auto-insurance telematics.

6.6 Section 6: Conclusion

The conclusion emphasizes that consumer welfare depends on both competition and the design of disclosure technology. Privacy policy should not only ask whether consumers can opt out, but what kinds of verifiable partial disclosure they can make and to whom.

7 Figures and detailed interpretation

7.1 Figure 1: Zeno's Partition, Prices, and Payoffs

Read:

- Panel (a) displays the recursive partition of consumer valuations: $[1/2, 1]$, $[1/4, 1/2]$, $[1/8, 1/4]$, and so on.
- Panel (b) plots the segment prices as a step function and consumer payoffs as the distance between the dashed 45-degree line and the price step.
- In each segment, the monopolist prices at the bottom of the segment; all consumers in the segment buy.
- The shaded areas show consumer surplus created by partial pooling within each segment.

Detailed interpretation: Zeno's partition is the canonical example of how rich evidence can help consumers even against a monopolist. The construction avoids full revelation, which would allow the monopolist to charge each type her exact willingness to pay. It also avoids full pooling, which excludes low-value consumers and leaves high-value consumers paying a uniform monopoly price. Instead, it creates a sequence of segments where low-end types make the segment price low and high-end types earn surplus. The reason this can be an equilibrium is that off-path beliefs deter consumers from deviating to other messages: a consumer who tries to send a different message may be treated as the highest type compatible with that message.

Economic message: Rich partial disclosure lets consumers design a beneficial form of personalized pricing. The seller still price discriminates, but the segmentation leaves consumer surplus inside each group.

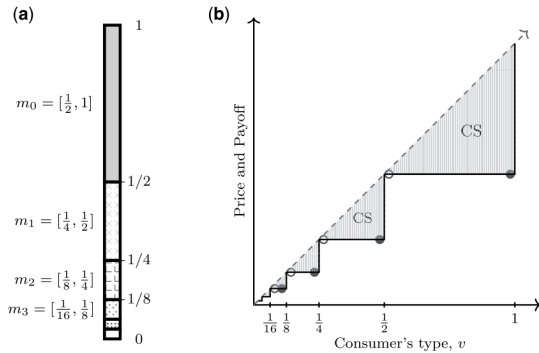


FIGURE 1

Figure 1: Zeno's Partition, Prices, and Payoffs

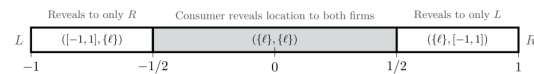


FIGURE 2

Figure 2: Disclosure Strategies for Every Type with Simple Evidence

7.2 Figure 2: Simple Evidence Disclosure Strategy in Competition

Read:

- The horizontal line is the consumer's location $\ell \in [-1, 1]$.
- Centrally located consumers reveal their location to both firms.
- Extreme right types reveal only to firm L , the distant firm, and conceal from firm R , the closer firm.
- Extreme left types reveal only to firm R , the distant firm, and conceal from firm L , the closer firm.

Detailed interpretation: The figure shows the core mechanism of the competitive model. Consumers near the middle are valuable to both firms, so full revelation makes the firms compete. Consumers near an extreme are naturally closer to one firm, so the close firm has market power. To reduce that market power, an extreme consumer reveals herself to the distant firm so that the distant firm can make a credible competitive offer, while concealing from the close firm so that the close firm cannot fully exploit her strong preference.

Economic message: Simple track/do-not-track evidence can be enough under competition because consumers can disclose asymmetrically across firms. Privacy control becomes a tool for creating outside options.

7.3 Figure 3: Rich Evidence Segmentation in Competition

Read:

- The figure partitions the line of consumer locations into multiple segments for the two firms.

- Some segments are large and shaded; some middle segments are omitted because the partition contains countably many intervals.
- The construction mirrors Zeno's partition but in a two-firm location setting.
- Each segment corresponds to a message that leads one firm to set a specific price while the rival often prices aggressively.

Detailed interpretation: Figure 3 shows how rich evidence can create a finer competitive disclosure scheme than Figure 2. Consumers reveal enough information to induce targeted offers, but not so much that the closer firm can extract all surplus. The recursive segmentation on both sides of zero balances two forces: providing enough information to make poaching profitable, and preserving enough pooling to keep prices low.

Economic message: Rich evidence refines competition by allowing more granular pooling and pricing. It illustrates that richer privacy dashboards can create welfare gains when consumers control which firms see which information.

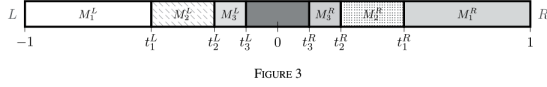


Figure 3: Segmentation Using Rich Evidence

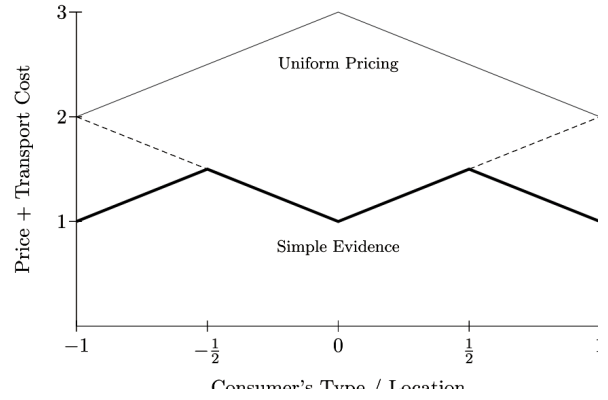


Figure 4: Interim Equilibrium Cost Under Uniform Pricing and Simple Evidence

7.4 Figure 4: Uniform Pricing vs. Simple Evidence Costs

Read:

- The vertical axis is total cost: price plus transport cost.
- The thin line shows total cost under uniform pricing without personalized pricing.
- The thick line shows total cost in the simple-evidence equilibrium.
- For the uniform distribution, simple evidence reduces expected cost by 50 percent.

Detailed interpretation: The figure makes the competitive welfare gain visually obvious. Uniform pricing creates high costs because each firm sets a price that reflects the full market rather than the specific competitive threat. Under simple evidence, consumers can manipulate the information seen by each firm. Centrally located types trigger strong competition; extreme types reveal to the distant firm to create a credible outside option. The thick line lies below the uniform pricing line for all types.

Economic message: Consumer-controlled disclosure can lower prices for every type in a competitive market. Personalized pricing is not necessarily exploitative when disclosure strengthens rivalry.

7.5 Figure 5: Real-World Verifiable Disclosure Interactions

Read:

- Panel (a) shows an EBT discount for Amazon Prime, an example of government-verified group pricing.
- Panels (b) and (c) show platform controls over interest-based ads and advertiser opt-outs.
- Panel (d) shows a firm-run verification mechanism: auto-insurance telematics that can justify a discount.
- The examples map to the paper’s theoretical categories: group verification, platform-mediated partial disclosure, and firm verification.

Detailed interpretation: Figure 5 shows that the paper’s evidence technologies are not merely abstract. EBT discounts resemble group pricing: a consumer verifies membership in a low-income group and receives a lower price. Platform dashboards resemble intermediary-administered message spaces: platforms decide what categories exist and whether consumers can remove them. Auto-insurance telematics resemble simple evidence in competitive markets: consumers can opt into monitoring to prove that they are lower risk. In each case, the administrator of evidence has real power because it determines the credible messages consumers can send.

Economic message: The design of verification institutions is central to privacy policy. The welfare effect of consumer control depends on who administers evidence, how fine the message space is, and whether disclosure is selective across firms.

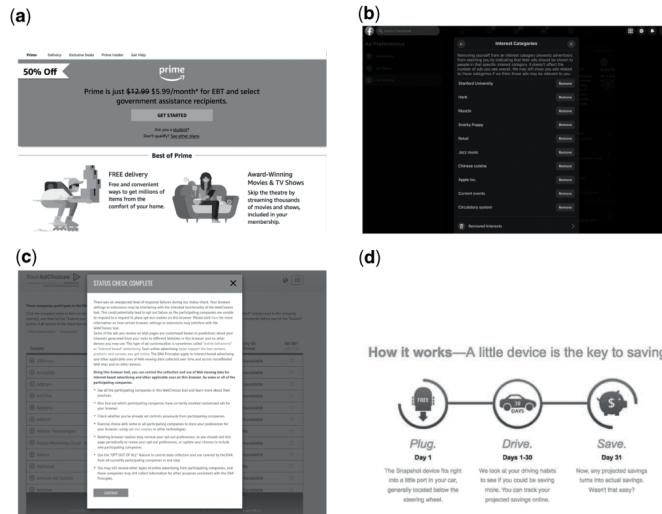


FIGURE 5

Figure 5: Screenshots from Examples of Verifiable Disclosure Interactions

8 Result map

- **Proposition 1:** simple evidence cannot improve monopoly outcomes. Full disclosure is too strong and non-disclosure is itself informative.
- **Proposition 2:** rich evidence can generate Pareto improvements under monopoly by enabling favorable pooling.
- **Proposition 3:** equilibria can be represented by efficient partitions without changing interim consumer surplus almost everywhere.

- **Proposition 4:** for power distributions, the greedy/Zeno segmentation is consumer optimal.
- **Proposition 5:** group-pricing evidence is enough for Pareto improvements; fully rich evidence is not necessary.
- **Proposition 6:** full revelation is an equilibrium in competitive markets.
- **Proposition 7:** simple evidence creates a welfare-improving selective-disclosure equilibrium under competition.
- **Proposition 8:** rich evidence creates an even more refined competitive segmentation equilibrium.
- **Proposition 9:** under symmetric log-concave densities, the constructed simple and rich evidence equilibria make every consumer type better off than uniform pricing.
- **Proposition 10:** under efficient passive equilibria, average consumer surplus is higher than under uniform pricing.

9 Short summary

- Firms increasingly use consumer data for personalized pricing, creating privacy and distributional concerns.
- The paper studies what happens when consumers control verifiable disclosure of their data.
- Consumer control does not automatically help because firms infer from both disclosure and non-disclosure.
- Under monopoly, simple evidence cannot improve consumer welfare.
- Rich evidence can help by allowing consumers to form price-reducing segments.
- Group-pricing evidence can also help, even if it is much coarser than fully rich evidence.
- Under competition, simple evidence can be powerful because consumers disclose selectively across firms.
- Disclosure can amplify competition by making distant firms credible poachers.
- In the uniform-location example, simple evidence reduces expected consumer costs by 50 percent.
- Real-world examples include EBT discounts, interest-based-ad opt-outs, and auto-insurance tracking programs.
- The paper’s policy message is that privacy rules must be designed around message spaces, verifiability, selectivity, and market structure.

10 Conclusion

10.1 Core conclusion

Ali, Lewis, and Vasserman show that consumer control over data can improve welfare, but only under the right disclosure technology and market structure. In monopoly, simple opt-in/opt-out evidence is generally too crude to help. Rich or group-based evidence is needed to induce favorable pooling and discounts. In competitive markets, by contrast, even simple evidence can improve welfare because consumers can use disclosure to intensify rivalry among firms.

10.2 Economic intuition

The economic intuition is that disclosure changes posterior beliefs, and posterior beliefs determine prices. Under monopoly, a seller uses information to extract surplus, so full revelation is dangerous. Consumers benefit only if disclosure lets them pool with other consumers in a way that lowers the segment price. Under

competition, disclosure can help by changing each firm’s perceived chance of winning the consumer. A consumer may disclose to a distant firm to create a credible outside option and conceal from the close firm to prevent exploitation.

10.3 Incremental contribution for privacy policy

The paper’s central policy contribution is that “consumer control” is not a single instrument. A track/do-not-track rule, a group-verification rule, a range-disclosure dashboard, and a platform-mediated opt-out system are different economic mechanisms. Their effects depend on the kinds of pooling and separation they allow. Good privacy design should ask: Can consumers disclose partially? Can they disclose selectively to one firm but not another? Can low-value or disadvantaged consumers verify group membership? Can platforms or governments credibly certify information?

10.4 Implications for firms and platforms

The results imply that platforms and firms may have incentives to design evidence systems that either help or hurt consumers. A platform that wants user engagement may design group categories that lower prices or improve matching. A firm facing competition may offer tracking to attract lower-cost consumers. A monopolist, however, may prefer evidence systems that preserve its ability to extract surplus. This makes the governance of disclosure administrators important.

10.5 Limitations

The model abstracts from privacy preferences, data breaches, endogenous data acquisition, algorithmic targeting errors, multi-product firms, and dynamic learning. It also treats evidence as verifiable and focuses on pricing rather than advertising persuasion. These limits are important for empirical applications, but they do not change the main theoretical message: the welfare effect of data control is determined by equilibrium inference and pricing.

10.6 Takeaway

The paper’s enduring message is that privacy and personalized pricing cannot be evaluated by asking whether more or less information is shared. The key question is who controls disclosure, what can be credibly disclosed, and how firms compete after seeing or not seeing the data. Consumer data rights are valuable when they allow consumers to create favorable pooling or stronger competition.